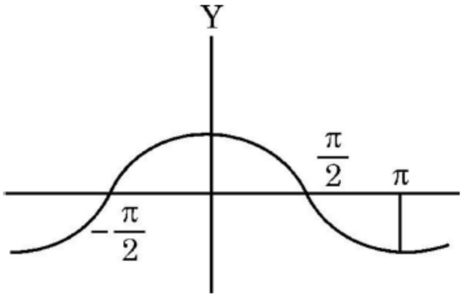
INVERSE TRIGONOMETRIC FUNCTIONS

1.	*Assertion (A) :* The range of the function $f(x) = 2\sin^{-1}x + \frac{3\pi}{2}$, where $x \in [-1, 1]$, is $\left[\frac{\pi}{2}, \frac{5\pi}{2}\right]$. *Reason (R) :* The range of the principal value branch of $\sin^{-1}(x)$ is $[0, \pi]$.
2.	Evaluate : $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) + \tan^{-1}(1)$
3.	**Assertion (A) :** Maximum value of $(\cos^{-1}x)^2$ is π^2. **Reason (R) :** Range of the principal value branch of $\cos^{-1}x$ is $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$
4.	Evaluate $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$.
5.	Draw the graph of $\cos^{-1}x$, where $x \in [-1, 0]$. Also, write its range.
6.	*Assertion (A) :* Range of $[\sin^{-1}x + 2\cos^{-1}x]$ is $[0, \pi]$. *Reason (R) :* Principal value branch of $\sin^{-1}x$ has range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
7.	Evaluate : $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}(0)$
8.	Draw the graph of $f(x) = \sin^{-1}x$, $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. Also, write range of f(x).
9.	*Assertion (A) :* All trigonometric functions have their inverses over their respective domains. *Reason (R) :* The inverse of $\tan^{-1}x$ exists for some $x \in \mathbb{R}$.

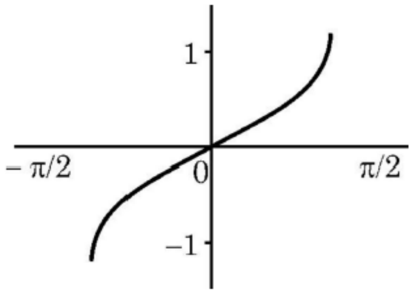
10.	Find the domain of $y = \sin^{-1}(x^2 - 4)$.
11.	Evaluate : $$\cos^{-1}\left[\cos\left(-\frac{7\pi}{3}\right)\right]$$
12.	$\sin\left[\frac{\pi}{3} + \sin^{-1}\left(\frac{1}{2}\right)\right]$ is equal to (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$
13.	Draw the graph of the principal branch of the function $f(x) =$ co
14.	*Assertion (A) :* $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$ is equal to $\frac{\pi}{6}$. *Reason (R) :* The range of the principal value branch of the function $y = \cos^{-1} x$ is $[0, \pi]$.
15.	Find the value of $\cos^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \text{cosec}^{-1}(-2$
16.	*Assertion (A) :* The principal value of $\cot^{-1}(\sqrt{3})$ is $\frac{\pi}{6}$. *Reason (R) :* Domain of $\cot^{-1} x$ is $\mathbb{R} - \{-1, 1\}$.
17.	Simplify : $$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$$
18.	Find the value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right.$

19.	Find the domain of the function $f(x) = \sin^{-1}(x^2 - 4)$. Also, find range.
20.	Evaluate : $$\cot^2\left\{\operatorname{cosec}^{-1} 3\right\} + \sin^2\left\{\cos^{-1}\left(\frac{1}{3}\right)\right\}$$
21.	Express $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$, where $\frac{-\pi}{2} < x < \frac{\pi}{2}$ in the simplest for
22.	Find the principal value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$.
23.	**Assertion (A)** : Domain of $y = \cos^{-1}(x)$ is $[-1, 1]$. **Reason (R)** : The range of the principal value branch of $y = \cos^{-1}(x)$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.
24.	Find value of k if $$\sin^{-1}\left[k \tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right] = \frac{\pi}{3}.$$

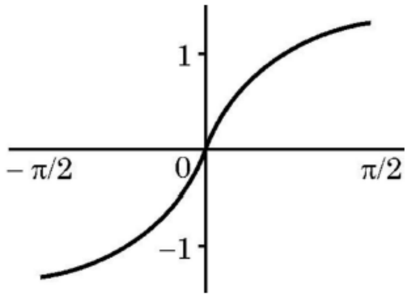
25. The graph of a trigonometric function is as shown. Which of the following will represent graph of its inverse ?



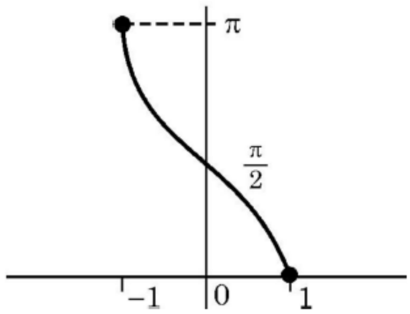
(A)



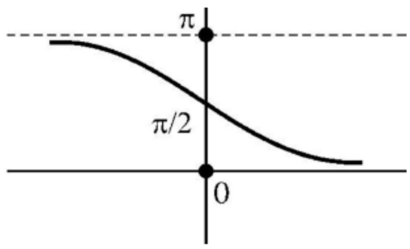
(B)



(C)



(D)



26.

$$\text{Evaluate : } \tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$$

27.

The principal value of $\sin^{-1}\left(\cos\frac{43\pi}{5}\right)$ is

(A) $\frac{-7\pi}{5}$

(B) $\frac{-\pi}{10}$

(C) $\frac{\pi}{10}$

(D) $\frac{3\pi}{5}$

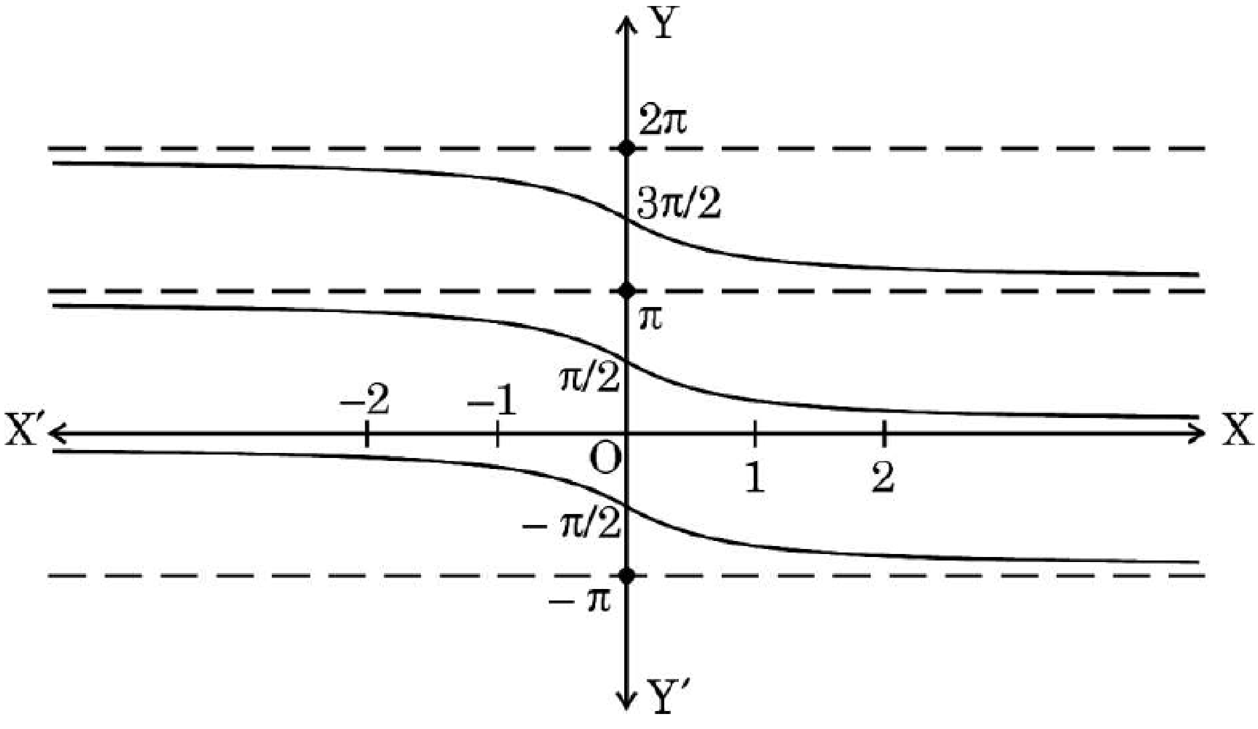
28. Domain of $\sin^{-1}(2x^2 - 3)$ is :

(A) $(-1, 0) \cup (1, \sqrt{2})$

(B) $(-\sqrt{2}, -1) \cup (0, 1)$

(C) $[-\sqrt{2}, -1] \cup [1, \sqrt{2}]$

(D) $(-\sqrt{2}, -1) \cup (1, \sqrt{2})$

29.	Simplify $\sin^{-1}\left(\dfrac{x}{\sqrt{1+x^2}}\right)$.
30.	Find domain of $\sin^{-1}\sqrt{x-1}$.
31.	The principal value of $\cot^{-1}\left(-\dfrac{1}{\sqrt{3}}\right)$ is : (A) $-\dfrac{\pi}{3}$ (B) $-\dfrac{2\pi}{3}$ (C) $\dfrac{\pi}{3}$ (D) $\dfrac{2\pi}{3}$
32.	Find the domain of the function $f(x) = \cos^{-1}(x^2 - 4)$.
33.	The graph shown below depicts :  (A) $y = \cot x$ (B) $y = \cot^{-1} x$ (C) $y = \tan x$ (D) $y = \tan^{-1} x$

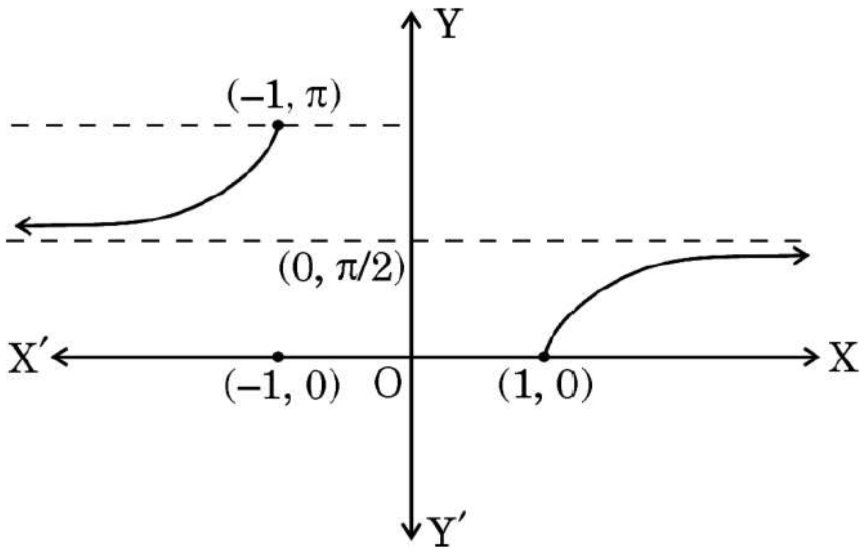
34. $\left[\sec^{-1}(-\sqrt{2})-\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right]$ is equal to :

(A) $\frac{11\pi}{12}$ (B) $\frac{5\pi}{12}$

(C) $-\frac{5\pi}{12}$ (D) $\frac{7\pi}{12}$

35. Find the domain of $f(x) = \sin^{-1}(-x^2)$.

36. The given graph illustrates :



(A) $y = \sec^{-1} x$ (B) $y = \cot^{-1} x$

(C) $y = \tan^{-1} x$ (D) $y = \mathrm{cosec}^{-1} x$

37. Domain of $f(x) = \cos^{-1} x + \sin x$ is :

(A) R (B) $(-1, 1)$

(C) $[-1, 1]$ (D) ϕ

38. *Assertion (A) :* Set of values of $\sec^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is a null set.

Reason (R) : $\sec^{-1} x$ is defined for $x \in R - (-1, 1)$.